

Physical Limitation of Signal Conversion

Physical limitations of signal conversion is crucial for Power Budgeting."

- the light signal becomes too weak due to fiber loss, the photocurrent will drop below the noise floor, leading to a high Bit Error Rate (BER).

In an optical communication system, the receiver's job is to convert light (photons) back into electricity (electrons).

1) Quantum Efficiency (η):
This represents the success rate of the photodiode. A 70% efficiency means that for every 100 photons that hit the detector, only 70 successfully generate an electron hole pair to create current.

Responsivity (R):
$$\frac{I_{ph}}{P_{in}} = \frac{P_{ph}}{P_{in}} \cdot \eta$$

This is a practical measure for engineers. While quantum efficiency is a ratio of particles, Responsivity tells us exactly how many Amperes of current we get for every watt of light power. It is wavelength-dependent because the energy of a photon changes with its color.

© Photocurrent (I_p):

This is the actual electrical signal that will be amplified and processed by the rest of the computer system to recover the transmitted data.

Ex: An optical uses a photodiode with a quantum efficiency of 70% at a wavelength of 1310 nm.

① Calculating Responsivity (R):

$$R = \frac{\eta q}{h\nu} = \frac{\eta q \lambda}{hc}$$

q = electron charge = 1.6×10^{-19} C here

h = Planck's constant = 6.626×10^{-34} Js $\eta = 70\% = 0.7$

c = Speed of light = 3×10^8 m/s $\lambda = 1310$ nm

$$R = \frac{0.7 \times 1.6 \times 10^{-19} \times 1310 \times 10^{-9}}{6.626 \times 10^{-34} \times 3 \times 10^8}$$

$$= 0.739 \text{ A/W}$$

② If the received optical power is $5 \mu W$

Photocurrent $I_p = R \times P_{in}$

$$I_p = 0.739 \times 5 \times 10^{-6}$$

$$I_p = 3.695 \times 10^{-6} A = 3.695 \mu A$$

Signal Dispersion

Signal dispersion is the spreading of light pulses as they travel, causing signal broadening and overlapping.

* Material Dispersion:

This occurs because the refractive index of glass varies with the wavelength of light.

↳ Since an LED light source isn't pure different colors travel at different speeds through the fiber.

* Pulse Broadening:

As these different speeds cause parts of the light pulse to arrive at different times, the pulse 'spreads out' or broadens as it travels.

* Intersymbol Interference (ISI):

If a pulse broadens too much, it starts overlapping with the pulse representing the next bit.

Ex: A glass fiber has a material dispersion (D_{mat}) of $80 \text{ ps/(nm}\cdot\text{km)}$ at an operating wavelength of 850 nm . The light source used is an LED with a spectral width ($\Delta\lambda$) of 40 nm .

- ① Calculate pulse broadening (ΔT) due to material dispersion for a fiber of 2 km length.

$$\Delta T = D_{mat} \times L \times \Delta\lambda$$

$$= 80 \times 2 \times 40$$

$$= 6400 \text{ ps}$$

$$= 6.4 \text{ ns (Ans)}$$

Here,

$$D_{mat} = 80$$

$$L = 2 \text{ km}$$

$$\Delta\lambda = 40 \text{ nm}$$

- ② Calculate maximum Bit Rate (R_B)

assuming the bit rate limited to $R_B \approx 1/(2\Delta T)$

using the provided approximation for a dispersion-limited link.

$$R_B = \frac{1}{2\Delta T}$$

$$= \frac{1}{2 \times 6.4 \times 10^{-9}} = 78125,000 \text{ bps}$$
$$= 78.13 \text{ Mbps.}$$

Requirement of Transmitter

The requirements for an optical fiber emitter are crucial for ensuring efficient and reliable optical communication.

Below are the key requirements:

Requirement	Description
Wavelength Compatibility	Must match the fiber's optical window to minimize attenuation and dispersion.
High Modulation speed	Must support rapid switching for high data rates; lasers are faster than LED for this purpose.
High Output Power	Sufficient power is required for this long-distance transmission with minimal signal degradation.
Low Spectral Width	Narrow spectral width reduces chromatic dispersion.
Temperature stability	Must maintain performance across temperature variations.
Reliability and Longevity	Should have a long operational life and low failure rates to minimize system downtime.
Efficient Coupling to fiber	Must efficiently couple light into the optical fiber core.
Low Noise	Should have minimal noise to maintain a high signal-to-noise (SNR) ratio in the communication system.
Cost-Effectiveness	Cost is important for short-distance, low-cost application.

(C) WDM - Wavelength Division Multiplexing

In WDM networks, multiple data signals are sent simultaneously over a single optical fiber by using different wavelengths of laser light.

To make this work, the hardware - transmitters, receivers and amplifiers - must meet specific high-performance requirements.

1. Transmitter Requirements:

① Precision wavelength: The laser must emit a very specific wavelength to avoid interfering with adjacent channels.

② Stability: The wavelength must remain constant regardless of temperature changes.

③ Narrow Linewidth: The light beam must be pure so the signal doesn't spread out over long distances.

2. Receiver Requirements:

- ① Selectivity: The receiver must be able to pick out one specific wavelength from the many arriving at the end of the fiber.
- ② Sensitivity: It must be able to detect very weak signals after they have traveled hundreds of km.
- ③ Wide Bandwidth: It needs to handle the high data rates sent by the transmitters.

3. Optical Amplifiers:

- In the past, to boost a signal, we had to convert light to electricity, amplify it, and turn it back to light.
- Modern optical amplifiers boost the light signal directly.

① SOA - (Semiconductor Optical Amplifier):

- ↳ Mechanism - Uses a semiconductor chip.
- ↳ Pros - Small, can be integrated on circuits.
- ↳ Cons - Higher noise and lower power than fiber-based amplifiers.

② EDFA - (Erbium-Doped Fiber Amplifier):

- ↳ Mechanism - A piece of glass fiber is doped with Erbium ions.
- ↳ Standard - This is the gold standard for long-distance internet because it can amplify many WDM channels at once without distortion.

(c) EDFA - (Erbium-Bismuth Fiber Amplifier)

→ Evolution - Similar to EDFA but uses Bismuth-doped glass to provide a wider bandwidth.

allowing even more WDM channels to be packed into the fiber.

Why use?

Wavelength Locker:

Wavelength locker is an important device in an optical communication system, that ~~use~~ helps to stable the fixed laser wavelength.

~~The~~ light come out from laser.

The wavelength of the light emitted from the laser can sometimes change due to temperature or other reasons. But in fiber optic communication a specific wavelength is very important.

The wavelength locker works to keep this wavelength fixed.

Tunable Laser:

A tunable laser can change its operating wavelength. This is useful because one backup laser can replace any failed laser in a WDM system.

Types:

① External Cavity Lasers (ECL):

Uses a physical grating that moves to change the color.

② Distributed Bragg Reflector (DBR):

Uses electrical currents to change the refractive index of a built-in grating, changing the wavelength.

③ Vertical Cavity Surface Emitting Laser (VCSEL):

Small and cheap they tune by moving a tiny top mirror using MEMS technology.

⇒ MEMS → (Micro-Electro-Mechanical System):

MEMS is a technology where very small mechanical + electrical components are built together on a chip.

(1) Ex: WDM Power Calculation

A WDM system has 8 channels. Each channel enters an EDFA with a power of -20 dBm .

EDFA has a gain of 25 dB .

① Calculate output power of a single channel in dBm

$$P_{\text{out}}(\text{dBm}) = P_{\text{in}}(\text{dBm}) + \text{Gain}(\text{dB})$$

$$= -20 + 25 = 5 \text{ dBm}$$

② Total output power in EDFA in mW .

First convert 5 dBm in mW :

$$P_{\text{mW}} = 10^{P_{\text{dBm}}/10} = 10^{5/10} = 3.16 \text{ mW}$$

Since there are 8 identical channels,

$$\begin{aligned} \text{Total Power} &= 8 \times 3.16 \\ &= 25.28 \text{ mW} \end{aligned}$$

WDM Receiver selectivity, Sensitivity,

Ans If increase WDM receiver selectivity then sensitivity not work properly then what effect does on system?

Passive Components

Passive components are devices that do not require electrical power to operate on the optical signal.

They function using the physical properties of light to direct, filter or modify the signal.

Passive Components

1. Mux & Demux: ✓

A mux combines multiple light signals of different wavelength into a signal fiber.

A demux does the opposite, splitting the combined signal back into individual wavelengths at the receiving end.

2. Optical switches:

These route an optical signal from one input fiber to one of several output fibers without converting the light to electricity.

3. Wavelength Converter: ✓

A device that changes the wavelength of an incoming signal to a different wavelength for the outgoing signal to avoid wavelength connection.

4. Couplers:

These split a ~~signal~~ optical single optical signal into two or more fiber or combine multiple signals into one.

5. Circulators & Isolators:

Isolator allows light to travel in only one direction, preventing reflection from damaging the laser.

Circulator is a multi-port device where light entering port 1 exits port 2 and light entering port 2 exits port 3.

6. Attenuator:

A device used to reduce power level of an optical signal if it is too strong for the receiver.

7. Optical filters:

These allow only a specific range of wavelength to pass through while blocking or reflecting others.

Ex: 1

Attenuation Problem.

current power : +3 dBm

Target power : -5 dBm

$$\text{Required reduction} = P_{in} - P_{max} = 3 - (-5) \\ = 8 \text{ dB}$$

An Attenuator of at least 8 dB is required.

$$0 \text{ dBm} \rightarrow 1 \text{ mW}$$

$$-10 \text{ dBm} \rightarrow 0.1 \text{ mW}$$

$$+10 \text{ dBm} \rightarrow 10 \text{ mW}$$

Ex: 2

A laser sends a signal with 10 mW of power into a 1:9 symmetric coupler.

① Calculate the power in mW at each output port.

Since it is a 1:9 symmetric split

$$P_{out} = \frac{10}{9} \text{ mW} = 2.5 \text{ mW}$$

② Output power of one port in dBm

$$P_{dBm} = 10 \log_{10}(P_{mW}) = 10 \log_{10}(2.5)$$

$$P_{dBm} = 3.98 \text{ dBm} \quad (h)$$

Optical Fiber Cable Design

Optical fiber cables are designed to protect fragile fibers maintain signal quality, and ensure mechanical strength.

Fiber Buffering:

Buffering is the first layer of protection applied to the fiber to prevent microbending and physical damage.

Types of Buffering:

① Tight Buffering: A thick layer of plastic is applied directly onto the fiber coating.

↳ It is used for indoor cables where flexibility and durability are needed.

② Loose-Tube Buffering: The fiber sits loosely inside a larger, hard plastic tube which is often filled with a water-resistant gel.

↳ This allows the fiber to move slightly when the cable expand or contracts due to temperature.

(44) Cable Strength and structural Members:

→ To prevent the fiber from being stretched or crushed during installation, strength members are added to the cable.

Mechanism:

① Central Strength Member:

A rod made of steel or fiberglass placed in the center of the cable to provide rigidity.

② Aramid Yarns (Kevlar):

strong synthetic fibers wrapped around the buffer tubes to provide high tensile strength, allowing the cable to be pulled through conduits without breaking.

Ex: Long-haul undersea cables use multiple layers of steel wire armor to withstand extreme ocean pressure.



DFB Laser Vs Fabry-Perot Laser (4 km)

⇒ Fabry-Perot (FP) Laser:

This laser uses two mirrors at the ends of the semiconductor material.

- Light bounces back and forth between them, but this often results in multiple modes or wavelengths being emitted.
- This causes more dispersion over long distances.

⇒ Distributed Feedback Laser:

Instead of simple mirrors, a DFB laser has a Bragg Grating etched directly into the semiconductor.

- This grating acts like a highly specific filter that only allows one precise wavelength to be amplified.

Comparison:

If an FP laser is like a flashlight emitting a broad beam, a DFB laser is like a high-precision pointer emitting a single, pure color.

This makes DFB lasers essential for high-speed WDM systems.

Fabrication of fiber Couplers:

Fiber couplers are passive components used to split or combine fiber signals.

① Fused Biconical Taper (FBT) method:

Process $\Rightarrow$ Two or more fibers are stripped of their coatings, placed side-by-side and twisted together.

Mechanism $\Rightarrow$ As the fiber cores get thinner, light leaks from one fiber into the other.

Ex: 3 dB Coupler created this way will take 10 mW of input power and split it into two 5 mW outputs.

② Polished Fiber Method:

Process $\Rightarrow$ Two fibers are partially embedded in separate glass blocks.

Mechanism $\Rightarrow$ Light moves between the fibers through "evanescent wave coupling".

Ex: This method is often used to create wavelength selective Couplers that can separate two different colors of light, such as 1310 nm & 1550 nm signals.

Optical Feedback & Laser Oscillation

Mechanism of Optical Feedback:

1. A laser consists of an active medium placed between two mirrors, forming an optical cavity or resonator.
2. When light is emitted within the medium, it bounces back and forth between the mirrors. This process is known as optical feedback.
3. As the light passes through the medium repeatedly, it triggers stimulated emission which amplifies the signal.
4. Oscillation occurs when the gain within the medium exactly matches the losses. At this point, the device emits a steady coherent beam of laser light.

Spectral Output:

- This feedback mechanism provides a distinctive spectral output because the cavity only supports specific "resonant frequencies" that fit perfectly between the mirrors.
- A laser using optical feedback has a narrow linewidth, meaning the light beam is extremely pure and concentrated on specific colors.

Fusion Splicing:

Fusion splicing is a technique used to join two optical fibers permanently by using localized heat to melt the glass ends together.

Advantages:

- **Low loss:** - It provides the lowest signal loss of all joining methods, typically less than 0.1 dB .
- **High Reliability:** - The resulting joint is almost as strong as the original fiber and is highly stable against temperature changes.
- **No Reflection:** - Because the glass becomes one continuous piece, there is virtually no back-reflection that could damage sensitive layers.

Drawbacks

- High initial cost:- Requires expensive, specialized equipment
- Precision Required:- The fibers must be perfectly cleaned and cleaved before splicing, even a tiny speck of dust can ruin the joint.
- Fragility :- The spliced area is stripped of its protective coating and becomes brittle, requiring a specialized "protection sleeve" to be added afterward.

((Ex: A four port multimode fiber FBT coupler has
60 μ W optical power launched into port 1,
The measured output powers at port 2, 3 & 4
are 0.004, 26.0 and 27.5 μ W respectively.

Solu:

Given data for port 1:

Input power (P_1) = 60 μ W

Output port 2 (P_2) = 0.004 μ W (crosstalk/isolated port)

Output port 3 (P_3) = 26.0 μ W

Output port 4 (P_4) = 27.5 μ W

1. Excess loss: This measure the power lost inside the coupler itself.

$$\begin{aligned} L_{\text{Exce}} &= 10 \log_{10} [P_1 / (P_3 + P_4)] \\ &= 10 \log_{10} [60 / (26 + 27.5)] \\ &= 0.50 \text{ dB} \end{aligned}$$

2. Insertion loss: Loss from input port to a specific output port.

$$\text{port 3: } 10 \log_{10} (60 / 26) = 3.63 \text{ dB}$$

$$\text{port 4: } 10 \log_{10} (60 / 27.5) = 3.38 \text{ dB}$$

3 Crosstalk: Measure how much light leaks into the isolated port.

$$\begin{aligned} \text{Formula} &= 10 \log_{10} (P_2/P_1) \\ &= 10 \log_{10} (0.009/60) \\ &= -41.76 \text{ dB} \end{aligned}$$

9. Split Ratio: The ratio of power between the two main output ports.

P_3 Ratio = $\frac{P_3}{P_3 + P_4} \times 100\% = \frac{26}{26 + 27.5} \times 100\%$
 $= \frac{26}{53.5} \times 100\%$
 $= 48.6\%$

P_q Ration = 51.4% (2A000) oblique position

4 Band Gap
— Describe how electrons move within a material to release energy.

① Direct Band Gap Semiconductor:

Def: The maximum energy level of the valence band and the minimum level of the conduction band occur at the same "momentum".

Result: Electrons can drop directly from the conduction band to the valence band.

usage: Used to make LEDs and lasers.

Ex: Gallium Arsenide (GaAs)

② Indirect Band Semiconductor:

Def: The conduction band minimum and valence band maximum occur at different momentum.

Result: An electron cannot drop directly; it must change its momentum, which usually releases energy as heat rather than light.

Usage: Generally poor for light emission but excellent for electronic switching

Ex: Silicon (Si) & Germanium (Ge)

5.1

P-n junction: A p-n junction is formed where a p-type semiconductor is joined with an n-type semiconductor.

Describe in general terms liquid-phase techniques for the preparation of multicomponent glasses for optical fiber?

Sol: Liquid-phase techniques involve the direct melting of high-purity raw materials to form a homogeneous multicomponent glass, which is later shaped into a preform for optical fiber drawing.

Process:

- High-purity oxides (SiO_2 , B_2O_3 , GeO_2 , P_2O_5) are selected
- Materials are melted at $1300-1500^\circ\text{C}$
- The molten glass is homogenized for uniform composition.
- Refining removes bubbles and impurities.
- Melt is cooled to form solid glass.

Feature :-

- Good compositional control
- Suitable for multicomponent glasses

Limitation:-

- Possible contamination and higher loss than vapor-phase methods.

(Ex) 5.1

An optical fiber has a core refractive index of 1.48. Two identical fibers with smooth, perpendicular end faces are connected, but a small air gap exists between them. The fiber axes are perfectly aligned.

(a) Calculate the Fresnel reflection coefficient at each interface.

$$F. R. C. = \left(\frac{n_1 - n_2}{n_1 + n_2} \right)^2$$

here $n_1 = 1.48$
 $n_2 = \text{air} = 1.0$

$$= \left(\frac{1.48 - 1}{1.48 + 1} \right)^2$$

$$= 0.0375$$

⑥ Determine the total transmission through the joint.

$$\begin{aligned}\text{Transmission at one interface } T_1 &= 1 - r \\ &= 1 - 0.0375 \\ &= 0.9625\end{aligned}$$

Since there are two interfaces:

$$\begin{aligned}T_{\text{total}} &= (1 - r)^2 \\ &= (0.9625)^2 \\ &= 0.9264 \quad (\text{m})\end{aligned}$$

⑦ Find the optical loss in dB at the joint due to the air gap.

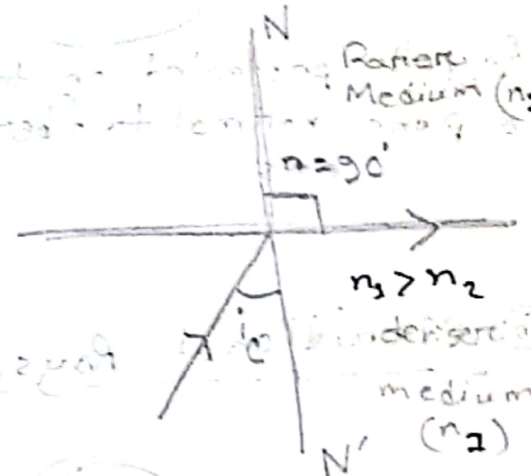
$$\begin{aligned}Loss_{\text{rec}} &= -10 \log_{10}(T_{\text{total}}) \\ &= -10 \log_{10}(0.9264) \\ &= 0.33 \text{ dB} \quad (\text{m})\end{aligned}$$

Some missing previous:

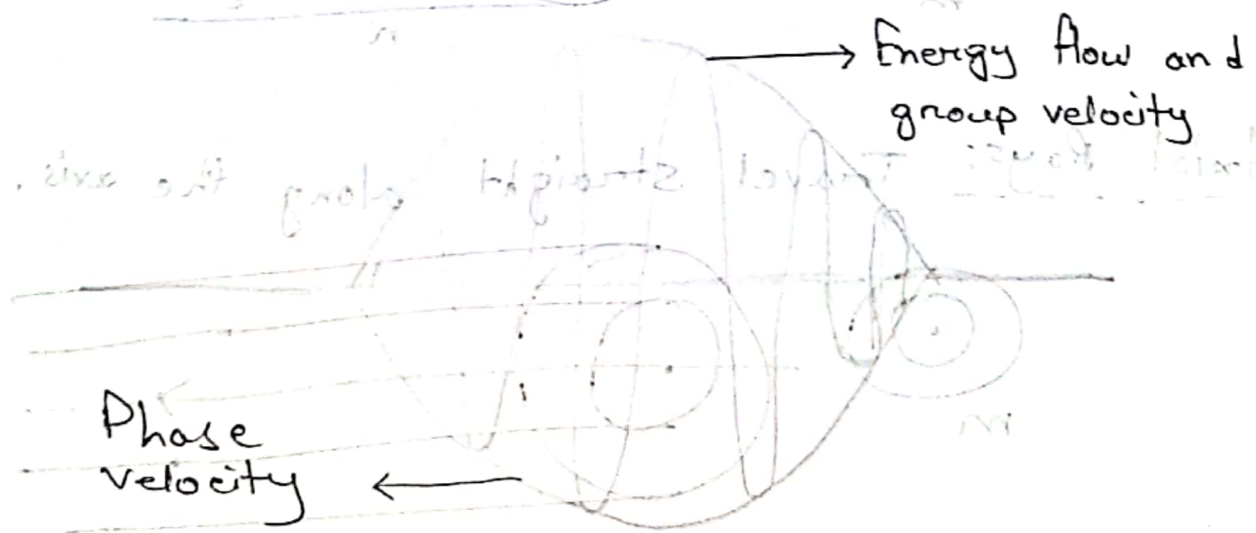
① Define the terms with suitable diagram: critical angle, Phase Velocity, Group Velocity.

① Critical angle: The angle of incidence for which the angle of refraction is 90 degree, is called as the critical angle.

$$\sin \phi_c = (n_2/n_1)$$



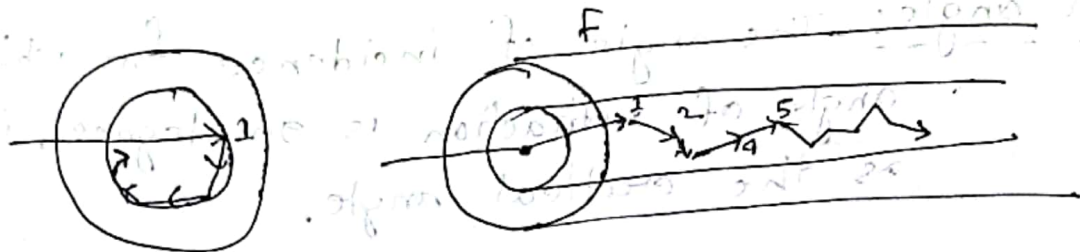
② Phase Velocity: The phase velocity refers to the velocity at which the phase of a wave moves while it travels through space or within a wave packet.



③ Group Velocity: The group velocity describes the velocity of the overall envelope shape of the wave packet.

The different Ray propagation:

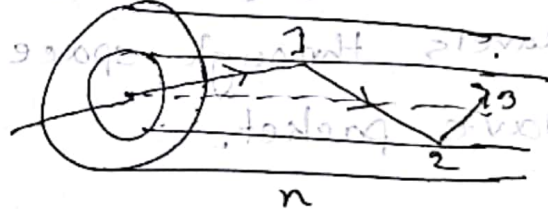
- ① Skew rays - don't cross the fiber axis but zig-zag around the axis.



Raypath projected on to a plane normal to fiber axis

$$(\text{m}) = \frac{\text{Fiber Length}}{n}$$

- ② Meridional Rays - always cross the fiber axis.



- ③ Axial Rays - Travel straight along the axis.

